EXP 19: Implementation of Naive Bayes classification for Bank Loan prediction

CODING:

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# Bank Loan Prediction using Naive Bayes

import math

# Dataset: [Age, Monthly Income (in thousands), Loan Status]
data = [
    [25, 20, "Rejected"],
    [30, 25, "Rejected"],
    [35, 30, "Approved"],
    [40, 50, "Approved"],
    [45, 60, "Approved"],
    [50, 55, "Approved"]
]

classes = ["Approved", "Rejected"]

# Calculate mean
def mean(values):
    return sum(values) / len(values)

# Calculate variance
def variance(values):
    m = mean(values)
    return sum((x - m) ** 2 for x in values) / len(values)
```

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# Gaussian probability
def gaussian(x, m, v):
    if v == 0:
        v = 0.0001

    return (1 / math.sqrt(2 * math.pi * v)) * math.exp(
        -((x - m) ** 2) / (2 * v)
    )

# Train the model
model = {}

for c in classes:
    rows = [row for row in data if row[2] == c]
    model[c] = []

    for i in range(2):
        values = [row[i] for row in rows]
        model[c].append((mean(values), variance(values)))

# Predict loan status
def predict(age, income):
    sample = [age, income]
    scores = {}

    for c in classes:
        rows = [row for row in data if row[2] == c]
        probability = len(rows) / len(data)
```

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    for i in range(2):
        m, v = model[c][i]
        probability *= gaussian(sample[i], m, v)

    scores[c] = probability

return max(scores, key=scores.get)

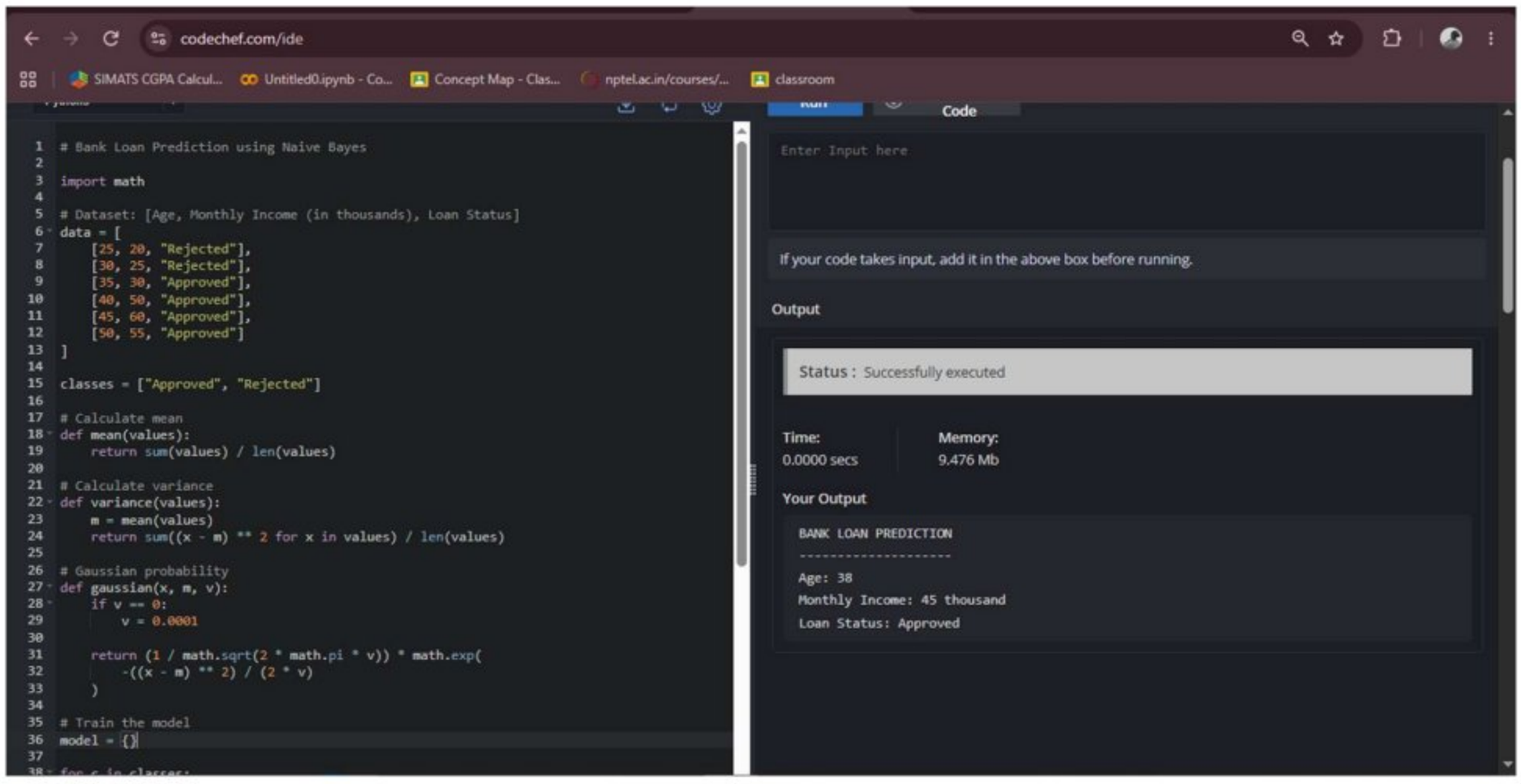
# Test data
age = 38
income = 45

result = predict(age, income)

print("BANK LOAN PREDICTION")
print("--------------------")
print("Age:", age)
print("Monthly Income:", income, "thousand")
print("Loan Status:", result)
```

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OUTPUT:



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